



Education

2021.09 - 2025.06	Shanghai Jiao Tong University	Energy and Power Engineering	B.Eng
2025.08 - present	Nanyang Technological University	Mechanical Engineering	M.Sc



Project Experiences

2025.9 ~ present Enhancing Fine-grained Manipulation of Humanoids Using VLA Model

■ Master's Thesis

Collected multimodal data using **MobileAloha** and developed a fine-tuning pipeline for VLA models for **fine-grained humanoid manipulation**. Reproduced and benchmarked **pi0**, **ACT**, and **Diffusion Policy** in the LeRobot framework, comparing their generalization, stability, transferability, and control responsiveness on real-world tasks.

2026.2 ~ present AI for Industry Challenge by Intrinsic (Part of Google)

■ International Challenge

Adapted Built an end-to-end hil-serl pipeline for 6-DoF robotic cable insertion based on the AIC benchmark, spanning training, inference, and evaluation. Reconstructed the AIC task in a **hil-serl** environment and developed the complete learning pipeline, including demonstration collection, reward classifier training, bc pretraining, **rlpd** policy optimization, **human-in-the-loop** refinement, and simulation evaluation.

2021.12 ~ 2023.8 Robomaster Competition (World Champion) Project Team Leader

■ National University Robot Competition

In 2022, I led the research, development, and maintenance of the infantry robot's mechanical structure. In 2023, I was responsible for the quadrotor aerial robot project. My work included **drone mechanical design** in SolidWorks, performing lightweight, aerodynamic, and vibration optimizations based on simulation and testing. I also conducted **visual positioning experiments** with the Intel RealSense T265, implementing sensor-fusion algorithms for localization and debugging **VIO/VINS**-based visual positioning performance.

2024.12 ~ 2025.5 Research on Flight Control of Kilogram Quadrotor Cyclocopter

■ Undergraduate Final Year Project

Development of flight control algorithm for quadrotor cyclocopter based on **PX4 platform**. Implemented **PID control algorithm** and sensor fusion **EKF algorithm** for the aircraft. Developed the **Gazebo plugin** to implement the **physical effect model** of the rolling wing and completed the **Gazebo simulation model of the aircraft**. Completed **sitl simulation** and **hitl simulation**, and achieved **hovering and attitude control** of the simulation model. Design the **control system hardware** and **electrical system** of the aircraft, and complete the hardware wiring.

2022.5 ~ 2023.9 Research on Modeling, Communication, and Control of SCARA Robot

■ Participation in Research Program (PRP)

Designed a **machine-vision-based screw pre-tightening system** for a SCARA robot. Trained a **YOLOv8** model to detect screws and their types. Developed a robot **end-effector** capable of automatic bit switching and implemented a **force-feedback control algorithm** to determine tightening levels. Performed forward/inverse kinematics analysis and completed robot modeling and simulation in **MATLAB**.



Internship

2024.6 ~ 2024.9 Shanghai Ranjing Technology Co., Ltd. Algorithm Engineer

Trained deep learning models and evaluated human skeleton keypoint recognition performance. Developed **rule-based algorithms** for multi-exercise repetition counting and violation detection. Designed image-processing methods for **standing long jump distance estimation**.



Academic Awards

2022	National University Robot Competition RoboMaster	National First Prize
2022	Shanghai Jiao Tong University C-class Scholarship	Scholarship
2023	National University Robot Competition RoboMaster	National First Prize
2024	Shanghai Jiao Tong University Academic Progress Scholarship	Scholarship
2025	China College Students' Mechanical Engineering Competition	National Third Prize
2025	NTU School of MAE Graduate Scholarship (~USD 40,000, top1%)	Scholarship